

# **GLASGOW COLLEGE UESTC**

## **Main Exam Paper**

### **Power Electronics (UESTC3022)**

### **Power Electronics (UESTCHN3014)**

**Date: (remember to complete when info available from Ruoli)**

**Time: (remember to complete when info available from Ruoli)**

**Attempt all PARTS. Total 100 marks**

**Use one answer sheet for each of the questions in this exam.**

**Show all work on the answer sheet.**

**For Multiple Choice Questions, use the dedicated answer sheet provided.**

**Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.**

**An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.**

**All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.**

**The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.**

**DATA/FORMULAE SHEET IS PROVIDED AT THE FRONT OF PAPER**

## Power Electronics 2 Formulae List

For an inductor:  $v_L(t) = L \frac{di_L(t)}{dt}$ ,  $i_L(t) = \frac{1}{L} \int v_L(t) dt$ ,  $E_L(t) = \frac{1}{2} L [i_L(t)]^2$

For a capacitor:  $v_C(t) = \frac{1}{C} \int i_C(t) dt$ ,  $i_C(t) = C \frac{dv_C(t)}{dt}$ ,  $E_C(t) = \frac{1}{2} C [v_C(t)]^2$

Average value:  $V_{ave} = \frac{1}{T} \int_0^T v(t) dt$

RMS value:  $V_{rms} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt}$

Duty cycle:  $D \equiv \phi = \frac{T_{on}}{T_{on} + T_{off}} = \frac{T_{on}}{T_{period}}$

Amplitude modulation index of SPWM:

$$m_a = \frac{\hat{V}_{control}}{\hat{V}_{carrier}}$$

Frequency modulation index of SPWM:

$$m_f = \frac{f_{carrier}}{f_{control}}$$

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} = i^2(t)R = v(t) \times i(t) = \text{force} \times \text{velocity} = \text{torque} \times \text{angular velocity}$$

Average power:

$$P = \int_0^T v(t)i(t)dt = \frac{V_{rms}^2}{R} = I_{rms}^2 R$$

reverse recovery power loss in a diode:

$$P_{RR} = Q_{RR} \times V_{R(blocking)} \times f_{switching}$$

MOSFET switching loss:

$$P_{switching} = \frac{f_{switching} V_{DS(off)}}{2} (T_{on} I_{on} + T_{off} I_{off})$$

Thermal circuit:

$$\Delta T = P_{device} \times \Sigma \theta_{XY} \quad \text{or} \quad \Delta T(^{\circ}\text{C}) = \frac{\text{power} \times \text{time}}{\text{thermal capacitance}}$$

Junction temperature of device:

$$T_{junction} = T_{sink} + P_{device} (\theta_{junction-case} + \theta_{case-sink})$$

Calculus:

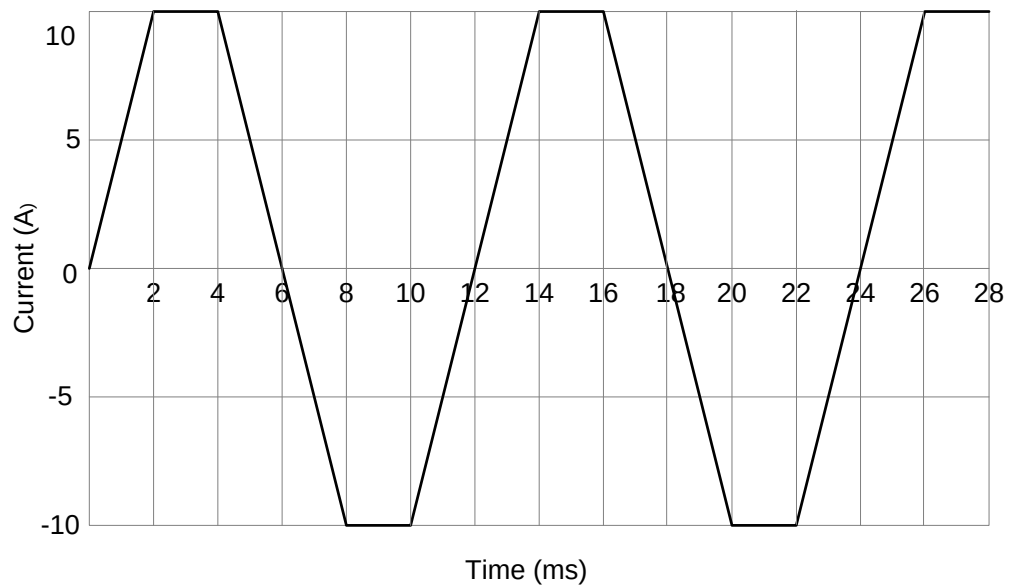
$$\cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\int \cos(a\theta) d\theta = \frac{1}{a} \sin \theta + c$$

Q1. Answer all questions in this section:

a. For the periodic load current waveform of Figure Q1-a, calculate:

- i. The average current. [1]
- ii. The RMS current. [2]
- iii. The form factor of the waveform. [2]



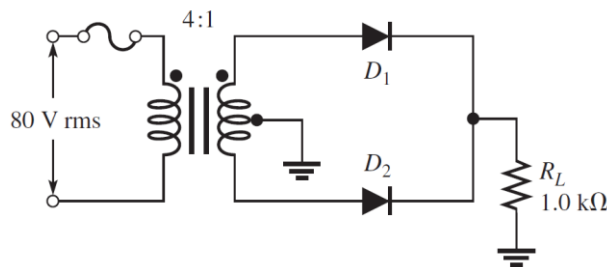
**Figure Q1-a**

b. Please answer the following questions:

- i. Draw and label a circuit to show how two silicon controlled rectifiers (SCRs) can be used to control a resistive load from an AC supply, allowing current to flow in both directions through the load. [2]
  - ii. Draw and label sketches of the current through the load to show how phase-angle control of SCRs is used to provide an average power:  
(1) less than 50% of its maximum value. [1.5]  
(2) above 50% of its maximum value. [1.5]
- c. Please list three advantages and three disadvantages of metal-oxide-semiconductor field-effect transistors (MOSFETs). [5]
- d. A switch-mode power supply contains two power semiconductor devices:
- A MOSFET switch, which dissipates 1.0 W.
  - A diode, which dissipates 2.0 W.

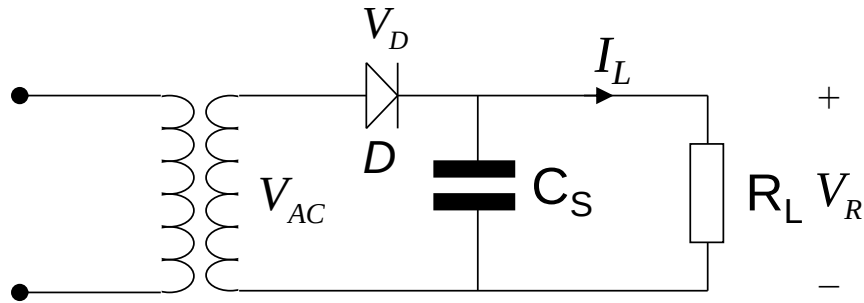
Both are in TO220 packages, which have thermal resistances  $\theta_{jc} = 3 \text{ } ^\circ\text{C/W}$  and  $\theta_{ca} = 60 \text{ } ^\circ\text{C/W}$ . The temperature of their junctions should not exceed  $125^\circ\text{C}$  and the ambient temperature is  $25^\circ\text{C}$ .

- i. Is it possible to use either device safely without a heatsink? [2]
  - ii. Despite the results of the previous analysis, the designer decides to mount both devices on a common heatsink. Specify the heatsink. (Hint: which component is most at risk from overheating?) [3]
- e. Consider the circuit in Figure Q1-e. Assume the power diode's forward voltage drop =  $0.7 \text{ V}$ . When an  $80 \text{ V}_{rms}$  sine voltage is applied to the primary winding of the transformer,
- i. What type of circuit is this? [1]
  - ii. What is the total peak secondary voltage ( $V_{p(sec)}$ )? [1]
  - iii. Draw the voltage waveform across  $R_L$  [1]
  - iv. What is the peak current through each diode? [1]
  - v. Determine the peak inverse voltage (PIV) for each diode. [1]



**Figure Q1-e**

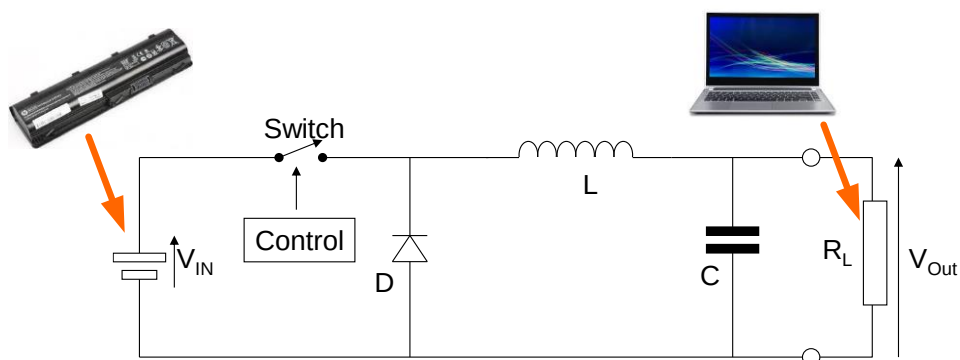
- Q2. For the half-wave diode rectifier circuit shown in Figure Q2, assuming the input voltage at the secondary side of the transformer is  $V_{AC} = 10\sin(100\pi t)$ , the capacitance  $C_s$  is  $20000 \text{ } \mu\text{F}$ , and the load resistance  $R_L$  is  $10 \text{ } \Omega$ .
- a. If the conduction angle of diode  $D$  is  $30^\circ$ , calculate the peak-to-peak ripple voltage in the output voltage  $V_R$  and the peak inverse voltage across the diode  $D$ . Assume the diode  $D$  is ideal for this question. [4]
  - b. If the output filter capacitor  $C_s$  is removed and the diode  $D$  has a conduction voltage drop of  $V_{D(on)} = 1 \text{ V}$ , sketch and label the voltage waveforms across the load  $R$  and the diode  $D$  separately. Clearly indicate the time scale and include the diode's voltage drop in your sketch. [6]



**Figure Q2**

- c. If the output filter capacitor  $C_S$  is removed, and the diode  $D$  in Figure Q2 is replaced with an SCR (Thyristor) with a delay angle  $\alpha=30^\circ$  and assume the SCR is ideal for this question.
- Briefly describe why SCRs are called half-controllable switches. [2]
  - Are SCRs unsuitable for use in systems that are purely DC? Why? [3]
  - Sketch the output voltage  $V_R$  waveform, clearly indicating the delay angle  $\alpha$  and the time scale. [4]
  - Derive and calculate the average and RMS values of the output voltage  $V_R$  respectively. [6]

Q3. (a) A switched-mode converter, shown in Figure Q3, is required to deliver 7.5 V at output power of 5 W from a 12.0 V supply. It runs in continuous conduction mode at 100 kHz. The circuit is shown in figure Q3 and has a 50  $\mu\text{H}$  inductor. You may assume that all components are ideal and that the input and output are smoothed so effectively that their voltages may be treated as constant.



**Figure Q 3**

- (i) What type of converter is it and how you identified it? [2]

- (ii) Derive the expression of the output voltage as a function of the input voltage. [6]
- (iii) Calculate the required duty cycle and ON time of the switching signal. [2]
- (iv) Calculate the average current passing through the inductor. [1]
- (v) Derive the mathematical expression of change in inductor current during switch close and switch open case. [6]
- (vi) Calculate the maximum and minimum inductor current. [3]

(b) Consider you need an output voltage of 24 V from a switch mode DC-DC converter. This converter will be used at different places where input DC voltages are different and can vary between 16 to 32 volts. Moreover, there is a requirement that input should be electrically isolated from the output. For this case, answer the following:

- (i) Which switch mode DC to DC converter will you use and why? [2]
- (ii) Which input parameter you will use to maintain the output to 24 V? Justify your answer with appropriate mathematical expression providing input-output voltage relationship? [3]

Q4. Figure Q4 shows a single-phase full bridge inverter.

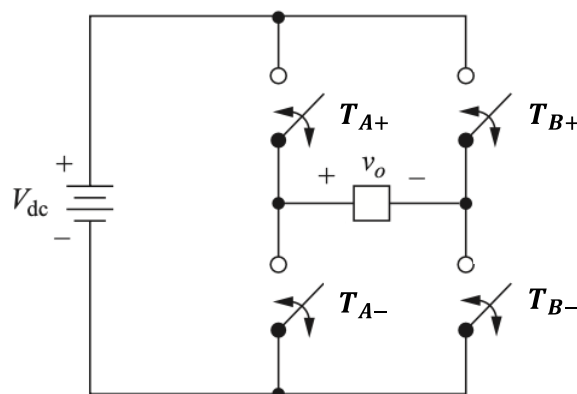


Figure Q4

The inverter is to be operated using PWM switching. Answer following:

- i. What are two types of PWM switching that can be applied to this inverter? [2]
- ii. Sketch the waveforms of the signals used in each type of PWM switching (identified in part (i)) to drive the switches. [5]

- iii. Write down the conditions for  $T_{A+}$ ,  $T_{A-}$ ,  $T_{B+}$  and  $T_{B-}$  to be in ON state, based on the values of signals identified in part(ii), for both types of PWM switching. [6]
- iv. Draw the diagram of circuit implementing the conditions identified in part (iii) for both types of switching. While making circuit diagrams, identify the inputs and outputs clearly. [6]
- v. Explain how the amplitude modulation ratio ( $m_a$ ) and frequency modulation ratio ( $m_f$ ) are related to the harmonics in the output of both types of the switching (consider  $m_a \leq 1$ ). [3]
- vi. Compare both PWM switching types in terms of performance and complexity of implementation. Build a scenario where you will prefer one type on another type. [3]